



# UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Fourth Semester – 2019

Paper: Statistics-IV

Course Code: STAT-203 / STT-22314 Part-I (Compulsory)

Time: 30 Min. Marks: 10

Roll No. in Fig. ....

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Signature of Supdt.: .....

**ATTEMPT THIS PAPER ON THIS QUESTION SHEET ONLY.**

**Division of marks is given in front of each question.**

**This Paper will be collected back after expiry of time limit mentioned above.**

**Q.1. Encircle the right answer cutting and overwriting is not allowed. (10x1=10)**

|        |                                                                                                                                                                                                                                                                                                                                                    |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (i)    | Suppose in a population of size N, the class frequencies of two attributes A and B are given by (A) and (B), Then we have the expectation of (A) and (B) combined is<br>(a) $\frac{(A)(B)}{N^4}$<br>(c) $\frac{(A)}{N^2} \cdot \frac{(B)}{N^2}$<br>(b) $\frac{(A)}{N} \cdot \frac{(B)}{N}$<br>(d) $\frac{(A)(B)}{N}$                               |
| (ii)   | In Duncan's multiple range test the "least significant range" is defined as<br>(a) $R_p = q_\alpha(p, v) \sqrt{S/r}$<br>(b) $R_p = q_\alpha(p, v) \sqrt{S^2/r^2}$<br>(c) $R_p = q_\alpha(p, v) \sqrt{S^2/r}$<br>(d) $R_p = q_\alpha(p, v) \sqrt{S/r^2}$                                                                                            |
| (iii)  | If the multiple correlation coefficient $R_{3.12} = 1$ , then it implies a<br>(a) Weak linear relationship<br>(b) Perfect relationship<br>(c) Perfect linear relationship<br>(d) High relationship                                                                                                                                                 |
| (iv)   | The second principle of an experimental design is<br>(a) Randomization<br>(b) Local control<br>(c) Replication<br>(d) None of these                                                                                                                                                                                                                |
| (v)    | The hypothesis is rejected at the $\alpha$ level of significance, when<br>(a) $F \leq F_{\alpha; (v_1, v_2)}$<br>(b) $F > F_{\alpha; (v_1, v_2)}$<br>(c) $F < F_{\alpha; (v_1, v_2)}$<br>(d) $F \geq F_{\alpha; (v_1, v_2)}$                                                                                                                       |
| (vi)   | Which of the following relationship holds<br>(a) $r_{12.3} = \frac{\sqrt{b_{12.3} \times b_{12.3}}}{\sqrt{b_{12.3} \times b_{12.3}}}$<br>(b) $r_{12.3} = \frac{\sqrt{b_{23.1} \times b_{32.1}}}{\sqrt{b_{23.1} \times b_{32.1}}}$<br>(c) $r_{13.2} = \frac{\sqrt{b_{13.2} \times b_{13.2}}}{\sqrt{b_{13.2} \times b_{13.2}}}$<br>(d) none of these |
| (vii)  | The range of multiple correlation coefficient is<br>(a) 0 to 1<br>(b) 0 to $+\infty$<br>(c) -1 to +1<br>(d) none of these                                                                                                                                                                                                                          |
| (viii) | If a chi-square test is performed on a contingency table with 5 rows and 4 columns. How many degrees of freedom should be used?<br>(a) 10<br>(b) 14<br>(c) 12<br>(d) 9                                                                                                                                                                             |
| (ix)   | If $s^2 = 1.57$ , $r = 4$ and $t_{0.025, (15)} = 2.13$ , then the "Least significant difference" is<br>(a) 1.98<br>(b) 1.97<br>(c) 1.79<br>(d) 1.89                                                                                                                                                                                                |
| (x)    | The standard error of estimate in multiple regression has degrees of freedom<br>(a) $k - n$<br>(b) $k - 1$<br>(c) $n - k$<br>(d) $n - 1$                                                                                                                                                                                                           |



**ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED**

**SECTION – II**

| Q. 2  | SHORT QUESTIONS                                                                                                                                                                                                 |     |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| (i)   | What is meant by a Two-way analysis of Variance and an interaction?                                                                                                                                             | (4) |
| (ii)  | Three variable have in pairs simple correlation coefficients given by $r_{12} = 0.8$ , $r_{13} = -0.7$ and $r_{23} = -0.9$ . Find the multiple correlation coefficient $R_{1.23}$ of $X_1$ on $X_2$ and $X_3$ . | (4) |
| (iii) | (a) Define a Contingency Table. How do you determine the number of degrees of freedom in an $r \times c$ contingency table.<br>(b) Discuss the important properties of chi-square distribution.                 | (4) |
| (iv)  | What is a multiple regression? Explain the basic difference between simple regression and multiple regression .                                                                                                 | (4) |
| (v)   | Define and discuss the use of randomization and replication in designing an experiment.                                                                                                                         | (4) |

**SECTION – III**

### LONG QUESTIONS

| Q.3             | <p>Five pennies were tossed 1000 times and the number of heads were observed as given below</p> <table border="1" style="margin: 10px auto; border-collapse: collapse; text-align: center;"> <tr> <td style="padding: 5px;">Number of heads</td><td style="padding: 5px;">0</td><td style="padding: 5px;">1</td><td style="padding: 5px;">2</td><td style="padding: 5px;">3</td><td style="padding: 5px;">4</td><td style="padding: 5px;">5</td></tr> <tr> <td style="padding: 5px;">Frequencies</td><td style="padding: 5px;">38</td><td style="padding: 5px;">144</td><td style="padding: 5px;">342</td><td style="padding: 5px;">287</td><td style="padding: 5px;">164</td><td style="padding: 5px;">25</td></tr> </table> <p>Test whether a binomial distribution gives a satisfactory fit to test data by applying the chi-square goodness-of-fit-test.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Number of heads | 0   | 1          | 2   | 3  | 4 | 5 | Frequencies | 38 | 144 | 342 | 287 | 164 | 25 | (6)    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|------------|-----|----|---|---|-------------|----|-----|-----|-----|-----|----|--------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|-----|
| Number of heads | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 1               | 2   | 3          | 4   | 5  |   |   |             |    |     |     |     |     |    |        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |
| Frequencies     | 38                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 144             | 342 | 287        | 164 | 25 |   |   |             |    |     |     |     |     |    |        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |
| Q.4             | <p>Perform the analysis of variance on the following data and analyse the treatment means using the “least significant difference” test with a 0.05 level of significance.</p> <table border="1" style="margin: 10px auto; border-collapse: collapse; text-align: center;"> <tr> <th colspan="2" rowspan="2"></th><th colspan="6">Treatments</th></tr> <tr> <th>1</th><th>2</th><th>3</th><th>4</th><th>5</th><th>6</th></tr> <tr> <td rowspan="4" style="padding: 5px;">Blocks</td><td style="padding: 5px;">1</td><td style="padding: 5px;">1</td><td style="padding: 5px;">3</td><td style="padding: 5px;">6</td><td style="padding: 5px;">4</td><td style="padding: 5px;">3</td><td style="padding: 5px;">2</td></tr> <tr> <td style="padding: 5px;">2</td><td style="padding: 5px;">1</td><td style="padding: 5px;">4</td><td style="padding: 5px;">4</td><td style="padding: 5px;">8</td><td style="padding: 5px;">5</td><td style="padding: 5px;">1</td></tr> <tr> <td style="padding: 5px;">3</td><td style="padding: 5px;">3</td><td style="padding: 5px;">6</td><td style="padding: 5px;">7</td><td style="padding: 5px;">8</td><td style="padding: 5px;">4</td><td style="padding: 5px;">3</td></tr> <tr> <td style="padding: 5px;">4</td><td style="padding: 5px;">2</td><td style="padding: 5px;">3</td><td style="padding: 5px;">2</td><td style="padding: 5px;">3</td><td style="padding: 5px;">2</td><td style="padding: 5px;">1</td></tr> </table> |                 |     | Treatments |     |    |   |   |             | 1  | 2   | 3   | 4   | 5   | 6  | Blocks | 1 | 1 | 3 | 6 | 4 | 3 | 2 | 2 | 1 | 4 | 4 | 8 | 5 | 1 | 3 | 3 | 6 | 7 | 8 | 4 | 3 | 4 | 2 | 3 | 2 | 3 | 2 | 1 | (6) |
|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                 |     | Treatments |     |    |   |   |             |    |     |     |     |     |    |        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |
|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 1               | 2   | 3          | 4   | 5  | 6 |   |             |    |     |     |     |     |    |        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |
| Blocks          | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 1               | 3   | 6          | 4   | 3  | 2 |   |             |    |     |     |     |     |    |        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |
|                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 1               | 4   | 4          | 8   | 5  | 1 |   |             |    |     |     |     |     |    |        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |
|                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 3               | 6   | 7          | 8   | 4  | 3 |   |             |    |     |     |     |     |    |        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |
|                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2               | 3   | 2          | 3   | 2  | 1 |   |             |    |     |     |     |     |    |        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |

P.T.O.

| Q.5    | <p>The following table shows the corresponding values of three variables <math>X_1</math>, <math>X_2</math> and <math>X_3</math>.</p> <table> <tr> <td><math>X_1</math></td><td>3</td><td>5</td><td>6</td><td>8</td><td>12</td><td>14</td></tr> <tr> <td><math>X_2</math></td><td>16</td><td>10</td><td>7</td><td>4</td><td>3</td><td>2</td></tr> <tr> <td><math>X_3</math></td><td>90</td><td>72</td><td>54</td><td>42</td><td>30</td><td>12</td></tr> </table> <p>a) Find the regression equation <math>X_3</math> on <math>X_1</math> and <math>X_2</math>.<br/> b) Estimate <math>X_3</math> when <math>X_1=10</math> and <math>X_2=6</math>.<br/> c) Compute <math>R_{3.12}</math> and <math>S_{3.12}</math></p> | $X_1$  | 3          | 5  | 6  | 8  | 12 | 14    | $X_2$ | 16 | 10 | 7  | 4  | 3     | 2  | $X_3$ | 90  | 72 | 54 | 42  | 30  | 12 | (6) |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|------------|----|----|----|----|-------|-------|----|----|----|----|-------|----|-------|-----|----|----|-----|-----|----|-----|
| $X_1$  | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 5      | 6          | 8  | 12 | 14 |    |       |       |    |    |    |    |       |    |       |     |    |    |     |     |    |     |
| $X_2$  | 16                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 10     | 7          | 4  | 3  | 2  |    |       |       |    |    |    |    |       |    |       |     |    |    |     |     |    |     |
| $X_3$  | 90                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 72     | 54         | 42 | 30 | 12 |    |       |       |    |    |    |    |       |    |       |     |    |    |     |     |    |     |
| Q.6    | <p>Given the following data</p> <table> <tr> <td>Y</td><td>2</td><td>5</td><td>7</td><td>8</td><td>5</td></tr> <tr> <td><math>X_1</math></td><td>8</td><td>8</td><td>6</td><td>5</td><td>3</td></tr> <tr> <td><math>X_2</math></td><td>0</td><td>1</td><td>1</td><td>3</td><td>4</td></tr> </table> <p>(a) Calculate the estimated regression equation, (i.e <math>Y=a+b_1X_1+b_2X_2</math>) for the above data.<br/> (b) State the meaning of the partial regression coefficients <math>b_1</math> and <math>b_2</math>.</p>                                                                                                                                                                                         | Y      | 2          | 5  | 7  | 8  | 5  | $X_1$ | 8     | 8  | 6  | 5  | 3  | $X_2$ | 0  | 1     | 1   | 3  | 4  | (6) |     |    |     |
| Y      | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 5      | 7          | 8  | 5  |    |    |       |       |    |    |    |    |       |    |       |     |    |    |     |     |    |     |
| $X_1$  | 8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 8      | 6          | 5  | 3  |    |    |       |       |    |    |    |    |       |    |       |     |    |    |     |     |    |     |
| $X_2$  | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 1      | 1          | 3  | 4  |    |    |       |       |    |    |    |    |       |    |       |     |    |    |     |     |    |     |
| Q.7    | <p>The following data obtained from a randomized complete block design with 3 treatments A, B and C and 3 blocks contain one missing observation represented by x.</p> <table> <tr> <th rowspan="2">Blocks</th><th colspan="3">Treatments</th></tr> <tr> <th>A</th><th>B</th><th>C</th></tr> <tr> <td>I</td><td>5</td><td>12</td><td>15</td></tr> <tr> <td>II</td><td>7</td><td>10</td><td>14</td></tr> <tr> <td>III</td><td>8</td><td>16</td><td>x</td></tr> </table> <p>Estimate the missing observation and prepare a table for analysis of variance.</p>                                                                                                                                                          | Blocks | Treatments |    |    | A  | B  | C     | I     | 5  | 12 | 15 | II | 7     | 10 | 14    | III | 8  | 16 | x   | (6) |    |     |
| Blocks | Treatments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |        |            |    |    |    |    |       |       |    |    |    |    |       |    |       |     |    |    |     |     |    |     |
|        | A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | B      | C          |    |    |    |    |       |       |    |    |    |    |       |    |       |     |    |    |     |     |    |     |
| I      | 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 12     | 15         |    |    |    |    |       |       |    |    |    |    |       |    |       |     |    |    |     |     |    |     |
| II     | 7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 10     | 14         |    |    |    |    |       |       |    |    |    |    |       |    |       |     |    |    |     |     |    |     |
| III    | 8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 16     | x          |    |    |    |    |       |       |    |    |    |    |       |    |       |     |    |    |     |     |    |     |